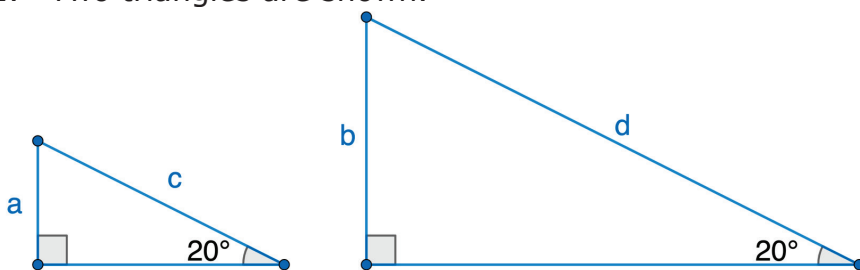




### 7.4.5 Wrap-Up: Ticket Out the Door

1. Two triangles are shown.



- a. Describe how the ratios  $\frac{a}{c}$  and  $\frac{b}{d}$  are related.
- b. What mathematical relationship(s) did you use to complete your description?

## Unit 7, Lesson 5: Using Trigonometric Ratios



### Benchmark

MA.912.T.1.2 Solve mathematical and real-world problems involving right triangles using trigonometric ratios and the Pythagorean Theorem.



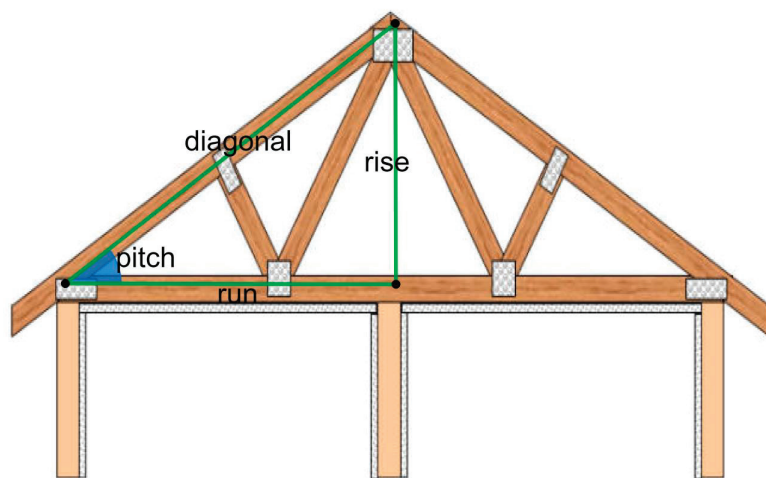
### Learning Targets

- ☐ I can use trigonometric ratios to find the side of a right triangle.
- ☐ I can use trigonometric ratios to find an angle of a right triangle.



### 7.5.1 Warm-Up: Notice and Wonder

In carpentry, right triangles are commonly used in constructing roofs. A roof's pitch is the steepness of the roof. It can be determined by the ratio between the two legs of the right triangle.

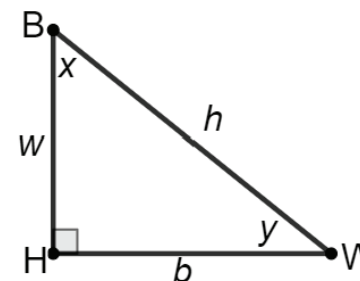


1. What do you notice or wonder about the use of right triangles in construction?



### 7.5.2 Collaboration: Using Trigonometric Ratios

1.  $\triangle BWH$  is shown, where  $h = 12.8$ ,  $w = 8$ ,  $x = 51.3^\circ$ , and  $y = 38.7^\circ$ .



- a. Select all of the equations that can be used to find the value of  $b$ .

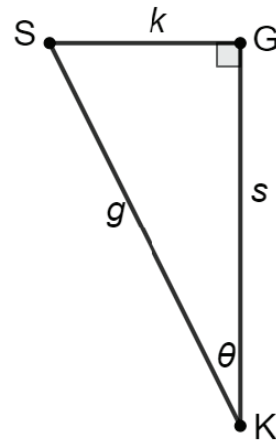
☐  $\cos(51.3^\circ) = \frac{8}{b}$       ☐  $\sin(51.3^\circ) = \frac{b}{12.8}$       ☐  $\tan(51.3^\circ) = \frac{b}{12.8}$

☐  $\cos(38.7^\circ) = \frac{b}{12.8}$       ☐  $\sin(38.7^\circ) = \frac{b}{12.8}$       ☐  $\tan(38.7^\circ) = \frac{8}{b}$

- b. Determine the value of  $b$ . Round to the nearest thousandth.

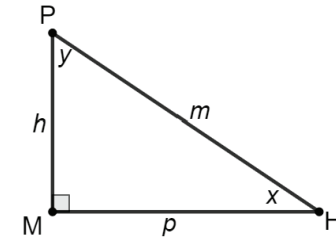
2.  $\triangle SGK$  is shown.

- a. Use a trigonometric ratio for  $\theta$  to write an equation that can be used to find the value of  $k$ .



- b. Which trigonometric ratio for  $\theta$  cannot be used to write an equation that can be used to find the value of  $k$ ?

3.  $\triangle PMH$  is shown. Work with your partner to determine which trigonometric ratio can be used to determine the unknown side or angle. Then, use the ratio to write an equation that can be used to determine the unknown. When writing the equation, use the variables  $h$ ,  $p$ ,  $m$ ,  $x$ , and  $y$ .



| Given                                          | Unknown         | Trigonometric Ratio                                                                         | Equation |
|------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------|----------|
| $m\angle PHM$ and length of $\overline{PM}$    | $\overline{PH}$ | <input type="radio"/> sine<br><input type="radio"/> cosine<br><input type="radio"/> tangent |          |
| $m\angle PHM$ and length of $\overline{PM}$    | $\overline{MH}$ | <input type="radio"/> sine<br><input type="radio"/> cosine<br><input type="radio"/> tangent |          |
| $m\angle HPM$ and length of $\overline{PM}$    | $\overline{PH}$ | <input type="radio"/> sine<br><input type="radio"/> cosine<br><input type="radio"/> tangent |          |
| Lengths of $\overline{MH}$ and $\overline{PH}$ | $m\angle HPM$   | <input type="radio"/> sine<br><input type="radio"/> cosine<br><input type="radio"/> tangent |          |
| Lengths of $\overline{MH}$ and $\overline{PM}$ | $m\angle PHM$   | <input type="radio"/> sine<br><input type="radio"/> cosine<br><input type="radio"/> tangent |          |



### 7.5.3 Guided Instruction: Using Inverse Trigonometric Functions to Find Angle Measure

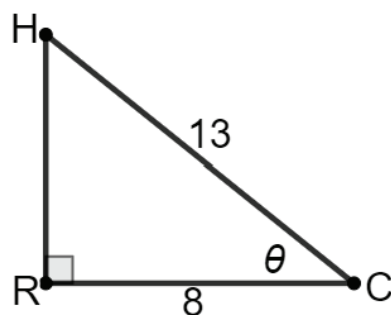


#### Inverse Trigonometric Functions

$\cos^{-1}\left(\frac{a}{c}\right)$  is arccosine.  $\sin^{-1}\left(\frac{b}{c}\right)$  is arcsine.  $\tan^{-1}\left(\frac{b}{a}\right)$  is arctangent. These functions can be used to determine the measure of an angle where  $a$  is the side adjacent the angle,  $b$  is the side opposite the angle, and  $c$  is the hypotenuse. The calculator should be in degree mode.

1. In  $\triangle HRC$ ,  $m\angle HRC = 90^\circ$ ,  $RC = 8$ , and  $HC = 13$ .

The equation  $\cos(\theta) = \frac{8}{13}$  can be used to determine  $\theta$ . To find  $\theta$  using a calculator use the  $\cos^{-1}\left(\frac{8}{13}\right)$ . Round any angle measurement below to the nearest whole degree.



- a. What is  $\cos^{-1}\left(\frac{8}{13}\right)$ ?

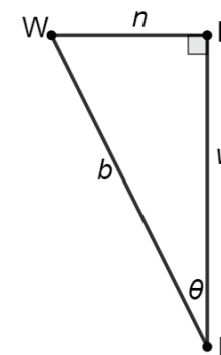
- b. Write an equation that can be used to find  $m\angle CHR$ .

- c. Find  $m\angle CHR$  using an inverse trigonometric function on a calculator. Be sure the calculator is in degree mode.

2.  $\triangle WLT$  is a right triangle with  $m\angle WTL = 90^\circ$ ,  $TL = 11$ , and  $WL = 23$ . What is the  $m\angle TWL$ ? Round to the nearest whole degree.

3.  $\triangle WBN$  is shown, where  $\angle B$  is a right angle.

- a. Determine the measure of  $\angle BNW$  if  $n = 36$  and  $w = 77$ .



- b. Gabriella used  $\sin^{-1}\left(\frac{77}{85}\right)$  to find the measure of  $\angle BNW$ . Determine if  $\sin^{-1}\left(\frac{77}{85}\right)$  results in the same degree measure as your answer.

- c. Discuss with your partner whether or not Gabriella's method is valid. Write a justification of your decision.



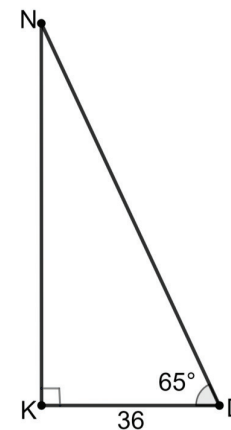
### 7.5.4 Guided Instruction: Using Trigonometric Ratios to Find Side Measures

1. In  $\triangle KDR$ ,  $m\angle DKR = 31.2^\circ$ ,  $m\angle KDR = 90^\circ$ , and  $KD = 17$ . What is the length of  $\overline{KR}$ ?

2.  $\triangle MAU$  is a right triangle with  $MA = 20.7$  and  $MU = 9.1$ . If  $m\angle MUA = 90^\circ$ , what is the  $m\angle AMU$ ?

3.  $\triangle NKD$  is shown, where  $\angle K$  is a right angle,  $KD = 36$ , and  $m\angle D = 65^\circ$ .

- a. Find the length of  $\overline{ND}$ . Show your work.



- b. Amelia and Henry found the length of  $\overline{NK}$  using two different methods. The first step of their work is shown. Complete their work.

| Amelia                     | Henry                            |
|----------------------------|----------------------------------|
| $36^2 + NK^2 = (85.183)^2$ | $\tan(65^\circ) = \frac{NK}{36}$ |

- c. Describe some possible reasons that Amelia's and Henry's answers are not exactly the same value.
- d. Whose method gives a precise answer?



### 7.5.5 Guided Practice: Using Trigonometric Ratios

1.  $\triangle ZME$  has  $ME = 14.9$  and  $m\angle ZEM = 41.5^\circ$ .
- a. If  $\angle ZME$  is a right angle, what is the length of  $\overline{ZM}$ ?

- b. Sharif used the information to determine the length of  $\overline{ZM}$ . Her work is shown below.

$$\tan 41.5^\circ = \frac{14.9}{ZM}$$

$$ZM \cdot \tan 41.5^\circ = 14.9$$

$$ZM = \frac{14.9}{\tan 41.5^\circ}$$

$$ZM \approx 16.841$$

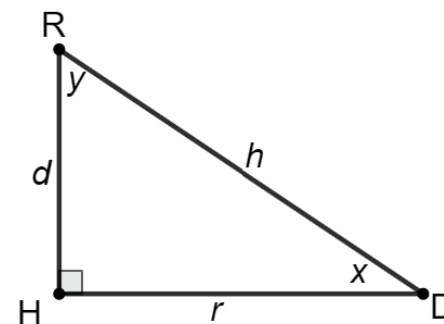
Find Sharif's error and correct it.

2.  $\triangle KNP$  is a right triangle with  $KP = 46$  and  $KN = 27.3$ . If  $\angle PKN$  is a right angle, what is the  $m\angle KPN$ ?



### 7.5.6 Wrap-Up: Ticket Out the Door

1.  $\triangle RDH$  is shown, where  $\angle RHD$  is a right angle.



- a. If  $d = 2.8$  and  $h = 4.6$ , what is the measure of  $y$ , in degrees?
- b. If  $x = 29^\circ$  and  $h = 14.5$  what is the value of  $d$ ?